DATA 480 - Predictive Analytics and Modeling

Description
Predictive Analytics & Modeling course is designed to build upon the foundation of the Predictive Analytics and Modeling Fundamentals. This course cover from intermediate to more advanced topics in predictive modeling, including unsupervised learning, ensemble learning, deep learning, reinforcement learning, and neural networks. Students will learn about the latest techniques and algorithms used in these areas and will gain hands-on experience using these methods to build predictive models. Students will learn how to analyze complex data sets and make predictions about future events. The course also covers important aspects of deployment, such as the ethics of AI, privacy and security, and explainability. Students will learn how to build responsible AI systems that take into account the ethical implications of AI and data privacy.

3 Credits

Time Guidelines
The standard instructional time for this course is 75 hours.

Prerequisites

- ARTI 406
- DATA 440
- STAT 400

Required Student Materials and Technology
Deep learning from scratch by Seth Weidman, ISBN 978-1-492-04141-2, O'Reilly Media

Course Assessment

Assignments	45%
Labs	30%
Project	25%
Total	100%

Use of Turnitin

The instructor may submit student work in this course to Turnitin's text-matching software program, to help assess the academic integrity of student work in this course. Turnitin results may be considered as one piece of evidence in academic misconduct hearings. Turnitin is an American company that keeps no unencrypted student identity data in the United States unless the students themselves choose to share this information in their submission. Students should limit their sharing of personally identifiable information by not including their names and student ID numbers within the text body of submitted assignments. SAIT has carefully reviewed this company's data management procedures.

Course Learning Outcomes

1. Implement Ensemble techniques

    Objectives:

    1.1 Define Ensemble Learning

    1.2 Explain different Ensemble techniques

    1.3 Identify different techniques to monitor Ensemble learning Models

    1.4 Demonstrate how to implement Ensemble methods

2. Create a solution using Unsupervised Learning

    Objectives:

    2.1 Explain different ways to choose a suitable clustering algorithm

    2.2 Evaluate the quality of clusters

    2.3 Identify the applications of clustering algorithms

    2.4 Demonstrate how to implement clustering algorithms

3. Demonstrate the use of Time Series models

    Objectives:

    3.1 Identify the components of time series analysis

    3.2 Identify the applications of time series analysis

    3.3 Explain the difference between stationary and non-stationary time series data types

    3.4 Implement different time series methods

4. Employ the fundamentals of Reinforcement learning

    Objectives:

    4.1 Describe Reinforcement Learning

    4.2 Explain different steps of Reinforcement Learning

    4.3 Identify the applications of Reinforcement Learning

    4.4 Implement Model free problem-solving methods

    4.5 Implement Model based problem solving methods

5. Summarize the fundamentals of Deep Learning

    Objectives:

    5.1 Explain the architecture of Neural Network

    5.2 Discuss Artificial Neural Network

    5.3 Outline the applications of Deep Neural Network

    5.4 Explain regulations and its type

    5.5 Analyze hyperparameter tuning and autoencoders

6. Develop a solution using Deep Learning Techniques

Objectives:

6.1 Explain Gradient Descent algorithm

6.2 Demonstrate how to implement Convolution Neural Network using convolution, pooling, and flattening techniques

6.3 Explain Recurrent Neural Network

6.4 Explain Transfer learning

6.5 Demonstrate how to Implement LSTM

7. Create an ethical AI implementation

Objectives:

7.1 Evaluate AI Governance

7.2 Explain Governance architecture for Human-centered AI

7.3 Identify different elements to ensure ethical AI

7.4 Discuss emerging rules and regulations

7.5 Demonstrate how to Implement ethical AI

SAIT Policies and Procedures:
Students should familiarize themselves with SAIT's Policies and Procedures, including but not limited to those in the Academic Student, Conduct, and Health, Safety, and Environment domains.